T.J.S. ENGINEERING COLLEGE

(AN AUTONOMOUS INSTITUTION)

Approved by AICTE, Affiliated to Anna University, Chennai.

TJS Nagar, Peruvoyal, Kavaraipettai, Chennai-601206.

B.E. MECHANICAL ENGINEERING

REGULATIONS 2026

CURRICULUM

(For the Students admitted from 2026-27)

T.J.S. ENGINEERING COLLEGE

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CHOICE BASED CREDIT SYSTEM (CBCS)

B.E. MECHANICAL ENGINEERING

Vision:

To provide quality education and focus on students to promote innovation leading to responsible engineers able to meet the emerging needs of the society as a whole.

Mission:

To impart quality education through outstanding, innovative academic practices.

To disseminate mechanical engineering knowledge for improving the quality of life.

To respond effectively to the changing needs of the industry and society.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs):

1. Effectuating success in careers by exploring with the design, digital and computational analysis of engineering systems, experimentation and testing, smart manufacturing, technical services, and research.

2. Amalgamating effectively with stakeholders to update and improve their core competencies and abilities to ethically compete in the ever-changing multicultural global enterprise.

3. To encourage multi-disciplinary research and development to foster advanced technology, and to nurture innovation and entrepreneurship in order to compete successfully in the global economy.

4. To globally share and apply technical knowledge to create new opportunities that proactively advances our society through team efforts and to solve various challenging technical, environmental and societal problems.

5. To create world class mechanical engineers capable of practice engineering ethically with a solid vision to become great leaders in academia, industries and society.

Program Outcomes:

1. Engineering Knowledge: Apply math, science, and engineering fundamentals to complex problems.

2. Problem Analysis: Identify and analyze complex problems using research and sustainability principles.

3. Design Solutions: Design systems and processes considering health, safety, cost, culture, and environment.

4. Investigations: Use experiments, modelling, and data analysis to reach valid conclusions.

5. Engineering Tools: Apply modern tools for modelling and problem-solving, recognizing their limits.

6. Society & Environment: Assess societal, legal, and environmental impacts of engineering solutions.

7. Ethics: Commit to ethics, human values, diversity, and legal compliance.

8. Teamwork: Work effectively as an individual and in multidisciplinary teams.

9. Communication: Communicate clearly in reports, presentations, and documentation across diverse groups.

10. Management & Finance: Apply management and economic principles in projects and teamwork.

11. Lifelong Learning: Engage in continuous learning, adapt to new technologies, and think critically.

PROGRAM SPECIFIC OUTCOMES:

PSO1 Design and develop efficient Mechanical systems in real time applications

PSO2 Utilize modern techniques and tools for addressing engineering challenges in the fields of Mechanical Engineering and providing sustainable innovation solutions.

PSO3 Demonstrate research aptitude, work in multidisciplinary teams and implement ethical practices.

Course Type	Definition
T	Theory
LIT	Lab Integrated with Theory
L	Laboratory

Period	Definition
L	Lecture Period
T	Tutorial Period
P	Practical Period

Course Category	Definition
HS	Humanities & Social Science Courses
BS	Basic Science Courses
ES	Engineering Science Courses
PC	Professional Core Courses
PE	Professional Elective Courses
OE	Open Elective Courses
EE	Employability Enhancement Courses
MC	Mandatory Courses

1 Hour Lecture (L) per week	1 Credit
1 Hour Tutorial (T) per week	1 Credit
1 Hour Practical (P) per week	0.5 Credit

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B.E. MECHANICAL ENGINEERING

CURRICULUM

SEMESTER I

S.NO	COURSE CODE	COURSE TITLE	Course Type	L-T-P	TCP	CREDIT	CATEGORY
1	26EN101	Professional English	T	2-0-0	2	2	HS
2	26TA101	Heritage of Tamils	T	1-0-0	1	1	HS
3	26MA101	Matrices and Calculus	T	3-1-0	4	4	BS
4	26PH201	Engineering Physics	LIT	2-0-2	4	3	BS
5	26CY201	Engineering Chemistry	LIT	2-0-2	4	3	BS
6	26CS201	Fundamentals of Programming in C	LIT	3-0-2	5	4	ES
7	26EN301	Life Skills for Engineers	L	0-0-2	2	1	HS
8	26ME301	Makerspace	L	0-0-4	4	2	ES
9		Physical Education-I	--	0-0-1	1	--	--
10		NCC/NSS/NSO/YRC	--	--	--	--	--
TOTAL					**27**	**20**	

SEMESTER II

S.NO	COURSE CODE	COURSE TITLE	Course Type	L-T-P	TCP	CREDIT	CATEGORY
1	26TA102	Tamils and Technology	T	1-0-0	1	1	HS
2	26MA102	Transforms and its Applications	T	3-1-0	4	4	BS
3	26PH101	Physics for Mechanical Engineering	T	3-0-0	3	3	BS
4	26CS202	Problem Solving using Python Programming	LIT	3-0-2	5	4	ES
5	26EE201	Basic Electrical and Electronics Engineering	LIT	3-0-2	5	4	ES
6	26ME201	Engineering Graphics	LIT	3-0-2	5	4	ES
7	26EN302	English Essentials	L	0-0-2	2	1	HS
8	26ME302	Re-Engineering for Innovation	L	0-0-4	4	2	ES
9	26SD301	Employability Enhancement Skills-I	L	0-0-2	2	1	EE
10		Physical Education-II	--	0-0-1	1	--	--
				TOTAL	**32**	**24**	

SEMESTER III

S.NO	COURSE CODE	COURSE TITLE	Course Type	L-T-P	TCP	CREDIT	CATEGORY
1	26MA107	Linear Algebra	T	3-1-0	4	4	BS
2	26ME102	Engineering Mechanics	T	3-1-0	4	4	PC
3	26ME103	Metallurgy and Material Science	T	3-0-0	3	3	PC
4	26ME104	Engineering Thermodynamics	T	3-0-0	3	3	PC
5	26ME202	Manufacturing Processes-I	LIT	3-0-2	5	4	PC
6	26ME203	Fluid Mechanics and Machinery	LIT	3-0-2	5	4	PC
7	26EN303	English Communication Skills Laboratory-I	L	0-0-2	2	1	HS
8	26SD302	Employability Enhancement Skills-II	L	0-0-2	2	1	EE
				TOTAL	**28**	**24**	

SEMESTER IV

S.NO	COURSE CODE	COURSE TITLE	Course Type	L-T-P	TCP	CREDIT	CATEGORY
1	26MA105	Statistics and Numerical Methods	T	3-1-0	4	4	BS
2	26EC104	Embedded Systems	T	3-0-0	3	3	ES
3	26ME204	Thermal Engineering	LIT	3-0-3	5	4	PC
4	26ME205	Theory of Machines	LIT	3-0-3	5	4	PC
5	26ME206	Manufacturing Processes-II	LIT	3-0-3	5	4	PC
6	26ME207	Strength of Materials	LIT	3-0-3	5	4	PC
7	26EN304	English Communication Skills Laboratory-II	L	0-0-2	2	1	HS
8	26SD303	Employability Enhancement Skills-III	L	0-0-2	2	1	EE
				TOTAL	**31**	**25**	

SEMESTER V

S.NO	COURSE CODE	COURSE TITLE	Course Type	L-T-P	TCP	CREDIT	CATEGORY
1	26CY101	Climate Change and Environmental Sustainability	T	2-0-0	2	2	HS
2	26ME105	Design of Machine Elements	T	3-1-0	4	4	PC
3		Open Elective-I	T/LIT			3	OE
4		Professional Elective-I	T/LIT			3	PE
5		Professional Elective-II	T/LIT			3	PE
6	26ME208	Metrology Measurements and Instrumentation	LIT	3-0-2	5	4	PC
7	26ME303	CAD/CAM Laboratory	L	0-0-4	4	2	PC
8	26SD304	Employability Enhancement Skills-IV	L	0-0-2	2	1	EE
9		Mandatory Course-I	T	2-0-0	2	--	MC
					TOTAL	**22**	
For Honours Degree							
1	26CD431	Capstone Design Project-Level I	CDP	0-0-12	12	6	
Or							
1		Honours Elective-I	T/LIT			3	
2		Honours Elective-II	T/LIT			3	
For Minor Degree							
1		Minor Elective-I	T/LIT			3	
2		Minor Elective-II	T/LIT			3	

SEMESTER VI

S.NO	COURSE CODE	COURSE TITLE	Course Type	L-T-P	TCP	CREDIT	CATEGORY
1	26ME106	Design of Transmission Systems	T	3-0-0	3	3	PC
2		Open Elective-II	T/LIT			3	OE
3		Professional Elective-III	T/LIT			3	PE
4		Professional Elective-IV	T/LIT			3	PE
5	26ME209	Heat and Mass Transfer	LIT	3-0-2	5	4	PC
6	26ME210	Finite Element Analysis	LIT	3-0-2	5	4	PC
7	26SD305	Employability Enhancement Skills-V	L	0-0-2	2	1	EE
8	26PW401	Engineering Design Project	L	0-0-4	4	2	EE
9		Mandatory Course-II	T	2-0-0	2	--	MC
					TOTAL	**23**	
For Honours Degree							
1	26CD432	Capstone Design Project-Level II	CDP	0-0-12	12	6	
Or							
1		Honours Elective-III	T/LIT			3	
2		Honours Elective-IV	T/LIT			3	
For Minor Degree							
1		Minor Elective-III	T/LIT			3	
2		Minor Elective-IV	T/LIT			3	

SEMESTER VII

S.NO	COURSE CODE	COURSE TITLE	Course Type	L-T-P	TCP	CREDIT	CATEGORY
1	26EN102	Human Values and Ethics	T	3-0-0	3	3	HS
2		Open Elective-III	T	3-0-0	3	3	OE
3		Professional Elective-V	T/LIT			3	PE
4		Professional Elective-VI	T/LIT			3	PE
5	26ME211	Mechatronics and Automation	LIT	3-0-2	5	4	PC
6	26IN401	Internship	--	--	--	1	EE
					TOTAL	**17**	
For Honours Degree							
1	26CD433	Capstone Design Project-Level III	CDP	0-0-12	12	6	
Or							
1		Honours Elective-V	T/LIT	3-0-0	3	3	
2		Honours Elective-VI	T/LIT	3-0-0	3	3	
For Minor Degree							
1		Minor Elective-V	T/LIT			3	
2		Minor Elective-VI	T/LIT			3	

SEMESTER VIII

S.NO	COURSE CODE	COURSE TITLE	Course Type	L-T-P	TCP	CREDIT	CATEGORY
1	26PW402	Project Work / Project Work-Cum-Internship	L	0-0-16	16	8	EE
				TOTAL	**16**	**8**	

Mandatory Courses-I							
S.NO	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**	**CATEGORY**
1	26MX101	Introduction to Women and Gender Studies	T	2-0-0	2	--	MC
2	26MX102	Elements of Literature	T	2-0-0	2	--	MC
3	26MX103	Film Appreciation	T	2-0-0	2	--	MC
4	26MX104	Disaster Risk Reduction and Management	T	2-0-0	2	--	MC

Mandatory Courses-II

S.NO	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**	**CATEGORY**
1	26MX105	Well Being with Traditional Practices	T	2-0-0	2	--	MC
2	26MX106	History of Science and Technology in India	T	2-0-0	2	--	MC
3	26MX107	Political and Economic Thought for a Humane Society	T	2-0-0	2	--	MC
4	26MX108	State, Nation Building and Politics in India	T	2-0-0	2	--	MC
5	26MX109	Industrial Safety	T	2-0-0	2	--	MC

Open Elective Courses-I							
S.NO	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**	**CATEGORY**
1	26CY102	Food Nutrition and Health	T	3-0-0	3	3	OE
2	26CY103	Environmental and Social Impact Assessment	T	3-0-0	3	3	OE
3	26MG116	Fintech Regulation	T	3-0-0	3	3	OE
4	26IT107	IT in Agricultural Systems	T	3-0-0	3	3	OE
5	26EN103	English for Competitive Examinations	T	3-0-0	3	3	OE
6	26CS103	Programming in Java	T	3-0-0	3	3	OE
7	26EC134	Digital Image Processing	T	3-0-0	3	3	OE
8	26AL103	Artificial Intelligence for Management	T	3-0-0	3	3	OE
9	26AL106	Foundations of Deep Learning	T	3-0-0	3	3	OE
10	26AL107	Fundamentals of Generative AI and Large Language Models	T	3-0-0	3	3	OE

Open Elective Courses-II							
S.NO	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**	**CATEGORY**
1	26MG118	NGOs and Sustainable Development	T	3-0-0	3	3	OE
2	26EN104	Democracy and Good Governance	T	3-0-0	3	3	OE
3	26CS104	Programming Data Structures and Algorithms using Python	T	3-0-0	3	3	OE
4	26AD102	Data Science for Engineers	T	3-0-0	3	3	OE
5	26AL105	AI in Human Resources Management	T	3-0-0	3	3	OE
6	26EC135	Remote Sensing	T	3-0-0	3	3	OE
7	26IT110	Privacy and Security in Social Media	T	3-0-0	3	3	OE
8	26EC140	Principles of Digital Communication	T	3-0-0	3	3	OE
9	26IT109	Information Theory	T	3-0-0	3	3	OE
10	26CS226	Digital Marketing	LIT	2-0-2	4	3	OE

Open Elective Courses-III							
S.NO	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**	**CATEGORY**
1	26EC136	Assistive Technology	T	3-0-0	3	3	OE
2	26MG119	Lean Concepts Tools and Practices	T	3-0-0	3	3	OE
3	26MG120	Production and Operations Management for Entrepreneurs	T	3-0-0	3	3	OE
4	26MG121	Cost Management of Engineering Projects	T	3-0-0	3	3	OE
5	26EE131	Batteries and Management System	T	3-0-0	3	3	OE
6	26EC137	Sensors and Actuators	T	3-0-0	3	3	OE
7	26EC138	Drone Technologies	T	3-0-0	3	3	OE
8	26EC139	Geographical Information Systems	T	3-0-0	3	3	OE
9	26IT106	Online Privacy	T	3-0-0	3	3	OE
10	26MG122	Patent Laws for Engineers and Scientists	T	3-0-0	3	3	OE

PROFESSIONAL ELECTIVES: VERTICALS

Vertical 1 Design	Vertical 2 Robotics and Automation	Vertical 3 Thermal Sciences	Vertical 4 Modern Mobility Systems	Vertical 5 Industrial Management	Vertical 6 Logistics and Supply Chain Management
Surface Engineering and Tribology	Sensors for Robotics	Renewable Energy Technologies	Advanced IC Engines	Engineering Ethics	Operations Research
Mechanical Vibration	Electrical Drives and Actuators for Robotics	Nuclear Engineering	Automobile Engineering	Industrial Engineering	Warehousing Automation
Design Concepts in Engineering	Industrial Robotics	Turbo Machinery Systems	Automotive Electrical and Electronics	Quality and Reliability Engineering	Material Handling Equipment, Repair and Maintenance
Design of Jigs and Fixtures	Robotics and Automation	Gas Dynamics and Jet Propulsion	Vehicle Body Engineering	Process Planning and Cost Estimation	Logistics in Manufacturing, Supply Chain and Distribution
Design of Pressure Vessels	Automation in Manufacturing	Refrigeration and Air Conditioning	Electric and Hybrid Vehicles	Industrial Safety and Maintenance	Transportation and Distribution Systems
Product Life Cycle Management	Digital Manufacturing	Computational Fluid Dynamics	Thermal Management of Batteries and Cells	Entrepreneurship Development	Inventory Control and Production Planning

PROFESSIONAL ELECTIVE COURSES - VERTICALS							
VERTICAL 1 - DESIGN							
S.NO	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**	**CATEGORY**
1	26ME108	Surface Engineering and Tribology	T	3-0-0	3	3	PE
2	26ME109	Mechanical Vibration	T	3-0-0	3	3	PE
3	26ME110	Design Concepts in Engineering	T	3-0-0	3	3	PE
4	26ME111	Design of Jigs and Fixtures	T	3-0-0	3	3	PE
5	26ME112	Design of Pressure Vessels	T	3-0-0	3	3	PE
6	26ME113	Product Life Cycle Management	T	3-0-0	3	3	PE

PROFESSIONAL ELECTIVE COURSES - VERTICALS							
VERTICAL 2 - ROBOTICS AND AUTOMATION							
S.NO	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**	**CATEGORY**
1	26ME167	Sensors for Robotics	T	3-0-0	3	3	PE
2	26ME168	Electrical Drives and Actuators for Robotics	T	3-0-0	3	3	PE
3	26ME114	Industrial Robotics	T	3-0-0	3	3	PE
4	26ME115	Robotics and Automation	T	3-0-0	3	3	PE
5	26ME116	Automation in Manufacturing	T	3-0-0	3	3	PE
6	26ME117	Digital Manufacturing	T	3-0-0	3	3	PE

PROFESSIONAL ELECTIVE COURSES - VERTICALS							
VERTICAL 3 - THERMAL SCIENCES							
S.No	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**	**CATEGORY**
1	26ME118	Renewable Energy Technologies	T	3-0-0	3	3	PE
2	26ME119	Nuclear Engineering	T	3-0-0	3	3	PE
3	26ME120	Turbo Machinery Systems	T	3-0-0	3	3	PE
4	26ME121	Gas Dynamics and Jet Propulsion	T	3-0-0	3	3	PE
5	26ME122	Refrigeration and Air Conditioning	T	3-0-0	3	3	PE
6	26ME123	Computational Fluid Dynamics	T	3-0-0	3	3	PE

PROFESSIONAL ELECTIVE COURSES - VERTICALS							
VERTICAL 4 - MODERN MOBILITY SYSTEMS							
S.No	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**	**CATEGORY**
1	26ME124	Advanced IC Engines	T	3-0-0	3	3	PE
2	26ME125	Automobile Engineering	T	3-0-0	3	3	PE
3	26ME126	Automotive Electrical and Electronics	T	3-0-0	3	3	PE
4	26ME127	Vehicle Body Engineering	T	3-0-0	3	3	PE
5	26ME128	Electric and Hybrid Vehicles	T	3-0-0	3	3	PE
6	26ME129	Thermal Management of Batteries and Cells	T	3-0-0	3	3	PE

PROFESSIONAL ELECTIVE COURSES - VERTICALS							
VERTICAL 5 - INDUSTRIAL MANAGEMENT							
S.No	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**	**CATEGORY**
1	26ME130	Engineering Ethics	T	3-0-0	3	3	PE
2	26ME131	Industrial Engineering	T	3-0-0	3	3	PE
3	26ME132	Quality and Reliability Engineering	T	3-0-0	3	3	PE
4	26ME133	Process Planning and Cost Estimation	T	3-0-0	3	3	PE
5	26ME134	Industrial Safety and Maintenance	T	3-0-0	3	3	PE
6	26ME135	Entrepreneurship Development	T	3-0-0	3	3	PE

PROFESSIONAL ELECTIVE COURSES - VERTICALS							
VERTICAL 6 - LOGISTICS AND SUPPLY CHAIN MANAGEMENT							
S.No	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**	**CATEGORY**
1	26MA108	Operations Research	T	3-0-0	3	3	PE
2	26ME137	Warehousing Automation	T	3-0-0	3	3	PE
3	26ME138	Material Handling Equipment, Repair and Maintenance	T	3-0-0	3	3	PE
4	26ME139	Logistics in Manufacturing, Supply Chain and Distribution	T	3-0-0	3	3	PE
5	26ME140	Transportation and Distribution Systems	T	3-0-0	3	3	PE
6	26ME141	Inventory Control and Production Planning	T	3-0-0	3	3	PE

MINOR ELECTIVES						
VERTICAL 1 - FINTECH AND BLOCKCHAIN						
S.No	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**
1	26MG101	Financial Management	T	3-0-0	3	3
2	26MG102	Fundamentals of Investment	T	3-0-0	3	3
3	26MG103	Banking Financial Services and Insurance	T	3-0-0	3	3
4	26IT103	Blockchain and its Applications	T	3-0-0	3	3
5	26IT104	Introduction to Fintech	T	3-0-0	3	3
6	26IT105	Fintech Personal Finance and Payments	T	3-0-0	3	3

MINOR ELECTIVES						
VERTICAL 2 - ENTREPRENEURSHIP						
S.No	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**
1	26MG123	Engineering Entrepreneurship and IPR	T	3-0-0	3	3
2	26MG105	Team Building and Leadership Management for Business	T	3-0-0	3	3
3	26MG106	Creativity & Innovation in Entrepreneurship	T	3-0-0	3	3
4	26MG107	Principles of Marketing Management for Business	T	3-0-0	3	3
5	26MG108	Human Resource Management for Entrepreneurs	T	3-0-0	3	3
6	26MG109	Financing New Business Ventures	T	3-0-0	3	3

MINOR ELECTIVES						
VERTICAL 3 - PUBLIC ADMINISTRATION						
S.No	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**
1	26MG110	Principles of Public Administration	T	3-0-0	3	3
2	26MG111	Constitution of India	T	3-0-0	3	3
3	26MG112	Public Personnel Administration	T	3-0-0	3	3
4	26MG113	Administrative Theories	T	3-0-0	3	3
5	26MG114	Indian Administrative System	T	3-0-0	3	3
6	26MG115	Public Policy Administration	T	3-0-0	3	3

MINOR ELECTIVES						
VERTICAL 4 - BUSINESS DATA ANALYTICS						
S.No	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**
1	26MA108	Statistics for Management	T	3-0-0	3	3
2	26AD101	Datamining for Business Intelligence	T	3-0-0	3	3
3	26AD104	Human Resource Analytics	T	3-0-0	3	3
4	26AD105	Marketing and Social Media Web Analytics	T	3-0-0	3	3
5	26AD106	Operation and Supply Chain Analytics	T	3-0-0	3	3
6	26AD107	Financial Analytics	T	3-0-0	3	3

VERTICAL 5 - SUSTAINABLE DEVELOPMENT						
S.No	**COURSE CODE**	**COURSE TITLE**	**Course Type**	**L-T-P**	**TCP**	**CREDIT**
1	26ME142	Sustainable Infrastructure Development	T	3-0-0	3	3
2	26ME143	Sustainable Agriculture and Environmental Management	T	3-0-0	3	3
3	26ME144	Sustainable Bio Materials	T	3-0-0	3	3
4	26ME145	Materials for Energy Sustainability	T	3-0-0	3	3
5	26ME146	Energy Efficiency for Sustainable Development	T	3-0-0	3	3
6	26ME147	Integrated Energy Planning for Sustainable Development	T	3-0-0	3	3

B.E. MECHANICAL ENGINEERING (R2026)

Course Category	**CREDITS PER SEMESTER**								**TOTAL CREDITS**
	I	**II**	**III**	**IV**	**V**	**VI**	**VII**	**VIII**	
HS	4	2	1	1	2		3		**13**
BS	10	7	4	4					**25**
ES	6	14		3					**23**
PC			18	16	10	11	4		**59**
PE					6	6	6		**18**
OE					3	3	3		**9**
EE		1	1	1	1	3	1	8	**16**
MC					x	x			**0**
	20	**24**	**24**	**25**	**22**	**23**	**17**	**8**	**163**